

FILE : CH8PP1

```
len = float(input("Enter length in cm: "))
if len < 0:
    print("Invalid input")
else:
    inch = len / 2.54
    print(len, "centimetres is equal to", inch, "inches")
```

FILE : CH8PP2

```
n = int(input("Enter number of items: "))
cost = 0
if n >= 100 :
    cost = n * 70
elif n >= 10 :
    cost = n * 100
else :
    cost = n * 120
print("Total Cost =", cost)
```

FILE : CH8PP3

```
hr = int(input("Enter hour between 1-12 : "))
n = int(input("How many hours ahead : "))
s = hr + n
if s > 12:
    s -= 12
print("Time at that time would be : ", s, "O'clock")
```

FILE : CH8PP4

```
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))
d = 0
if a > b :
    d = a - b
else :
    d = b - a
if d <= 0.001 :
    print("Close")
else :
    print("Not Close")
```

FILE : CH8PP5

```
year = int(input("Enter year: "))
if year % 400 == 0 :
    print(year, "is a Leap Year")
elif year % 100 == 0 :
    print(year, "is not a Leap Year")
elif year % 4 == 0 :
    print(year, "is a Leap Year")
else :
    print(year, "is not a Leap Year")
```

FILE : CH8PP6

```
a = int(input("Enter first side : "))
b = int(input("Enter second side : "))
c = int(input("Enter third side : "))
if a + b > c and b + c > a and a + c > b :
    print("Triangle Possible")
else :
    print("Triangle Not Possible")
```

FILE : CH8PP7

```
d = int(input("Enter a digit(0-9): "))
if d == 0 :
    print("Zero")
elif d == 1 :
    print("One")
elif d == 2 :
    print("Two")
elif d == 3 :
    print("Three")
elif d == 4 :
    print("Four")
elif d == 5 :
    print("Five")
elif d == 6 :
    print("Six")
```

```
elif d == 7 :
    print("Seven")
elif d == 8 :
    print("Eight")
elif d == 9 :
    print("Nine")
else :
    print("Invalid Digit")
```

FILE : CH8PP8

```
import math
n = int(input("Enter a number: "))
sr = math.sqrt(n);c = 0
for i in range(1, int(sr + 1)) :
    if (sr % i == 0) :
        c += 1
if c == 2 :
    print("Square root is prime")
else :
    print("Square root is not prime")
```

FILE : CH8PP9

```
n = int(input("Enter n: "))
x = n * 2 - 1
for i in range(x, 0, -2) :
    print(i)
```

FILE : CH8PP10

```
print("First Series:")
for i in range(1, 41, 3) :
    print(i, end = ' ')
print("\nSecond Series:")
x = 1
for i in range(1, 41, 3) :
    print(i * x, end = ' ')
    x *= -1
```

FILE : CH8PP11

```
sum = count = 0
print("Enter numbers")
print("(Enter 'q' to see the average)")
while True :
    n = input()
    if n == 'q' or n == 'Q' :
        break
    else :
        sum += int(n)
        count += 1
avg = sum / count
print("Average = ", avg)
```

FILE : CH8PP12

```
a = int(input("Enter first side : "))
b = int(input("Enter second side : "))
c = int(input("Enter third side : "))

if a == b and b == c :
    print("Equilateral Triangle")
elif a == b or b == c or c == a:
    print("Isosceles Triangle")
else :
    print("Scalene Triangle")
```

FILE : CH8PP13

```
a = int(input("Enter an integer: "))
if a % 10 == 4 :
    print("ends with 4")
elif a % 10 == 8 :
    print("ends with 8")
else :
    print("ends with neither")
```

FILE : CH8PP14

```
n = int(input("Enter a number greater than 20: "))
if n <= 20 :
    print("Invalid Input")
else :
    for i in range(11, n + 1) :
        print(i)
        if i % 3 == 0 and i % 7 == 0 :
            print("TopsyTopsy")
        elif i % 3 == 0 :
            print("Topsy")
        elif i % 7 == 0 :
            print("Topsy")
```

FILE : CH8PP15

```
print("Enter numbers:")
print("(Enter 'q/Q' to see the result)")
l = input()

if l != 'q' and l != 'Q' :
    l = int(l)
    while True:          # creates an infinite loop
        n = input()
        if n == 'q' or n == 'Q' :
            break
        n = int(n)
        if n > l :
            l = n
    print("Largest Number =", l)
```

FILE : CH8PP16

```
n = int(input("How many numbers you want to enter? "))
if n > 1 :
    l = int(input())
    sl = int(input())
```

```

if s1 > 1 :
    t = s1
    s1 = 1
    l = t
for i in range(n - 2) :
    a = int(input())
    if a > 1 :
        s1 = 1
        l = a
    elif a > s1 :
        s1 = a
print("Second Largest Number =", s1)
else :
    print("Please enter more than 1 number")

```

FILE : CH8PP17

```

print("Enter numbers:")
while True :
    n = input()
    n = int(n)
    t = n
    r = 0
    while (t != 0) :
        d = t % 10
        r = r * 10 + d
        t = t // 10
    if (n == r) :
        print(n, "is a Palindrome Number")
    else :
        print(n, "is not a Palindrome Number")

```

FILE : CH8PP18

```
x = int(input("Enter an integer: "))
temp = x
count = 0
digit = -1
while temp != 0 :
    digit = temp % 10
    count += 1
    temp = temp // 10
y = count * 10 + digit
print("Y =", y)
```

FILE : CH8PP19

```
m = int(input("Enter m: "))
n = int(input("Enter n: "))
for i in range(1, n) :
    if i % m == 0 :
        print(i, "is divisible by", m)
        if i % 2 == 0 :
            print(i, "is even")
        else :
            print(i, "is odd")
```

FILE : CH8PP20A

```
n = 2
d = 9
m = 1
sum = 0
for i in range(7) :
    t = n / d
    sum += t * m
    n += 3
    d += 4
    m *= -1
print("Sum =", sum)
```

FILE : CH8PP20B

```
n = int(input("Enter the value of n: "))
i = 1; sum = 0
while i <= n :
    sum += i ** 2
    i += 2
print("Sum =", sum)
```

FILE : CH8PP21

```
n = int(input("Enter the value of n: "))
sum = 0
for i in range(n + 1) :
    fact = 1
    for j in range(1, i) :
        fact *= j
    term = 1 / fact
    sum += term
print("Sum =", sum)
```

FILE : CH8PP22

```
n = int(input("Enter the value of n: "))
g1=g2=g3=0
for i in range(1, n + 1) :
    age = int(input("Enter employee age: "))

    if 26 <= age <= 35 :
        g1 += 1
    elif 36 <= age <= 45 :
        g2 += 1
    elif 46 <= age <= 55 :
        g3 += 1
print("Employees in age group 26 - 35: ", g1)
print("Employees in age group 36 - 45: ", g2)
print("Employees in age group 46 - 55: ", g3)
```

FILE : CH8PP23A

```
n = int(input("Enter the value of n: "))
g1=g2=g3=0
for i in range(1, n + 1) :
    age = int(input("Enter employee age: "))
    if 26 <= age <= 35 :
        g1 += 1
    elif 36 <= age <= 45 :
        g2 += 1
    elif 46 <= age <= 55 :
        g3 += 1
print("Employees in age group 26 - 35: ", g1)
print("Employees in age group 36 - 45: ", g2)
print("Employees in age group 46 - 55: ", g3)
```

FILE : CH8PP23B

```
x = int(input("Enter the value of x: "))
n = int(input("Enter the value of n: "));sum = 0
for i in range(1, n + 1) :
    term = x ** i / i
    sum += term
print("Sum =", sum)
```

FILE : CH8PP24

```
n=int(input("enter the N.O. of rows"))
for i in range(0,n):
    for j in range (0,i+1):
        if j== 0 or j==i:
            print("*",end="")
        else:
            print(" ",end="")
    print()
for i in range(n,-1,-1):
    for j in range(0,i+1):
        if j== 0 or j==i:
            print("*",end="")
        else:
            print(" ",end="")
    print()
```

FILE : CH8PP25

```
n=int(input("enter the no of rows :"))
for i in range(str(65),str(n)):
    for j in range(str(65),str(i)):
        print(j,end="")
    print()
```

FILE : CH8PP25C

```
n=int(input("enter the N.O. of rows"))
for i in range(0,n,2):
    for j in range (0,i+1,2):
        print(i,end=" ")
    print()
```

FILE : CH8PP25D

```
n=int(input("enter the N.O. of rows"))
for i in range(2,n,2):
    for j in range (2,i+1,2):
        print(i,end=" ")
    print()
```

FILE : CH8PP26

```
n=int(input("enter the N.O. of rows"))
for i in range(2,n,2):
    for j in range (2,i+1,2):
        print(i,end=" ")
    print()
```

FILE : CH8PP26A

```
name="AABCDEF"
for ch in range(0,len(name)) :
    for a in range(0,int(ch)):
        print(name[ch],end='')
    print()
```

FILE : CH8PP27

to find the largest, middle and smallest among 3 numbers

```
num1=float(input("Enter the 1st number: "))
num2=float(input("Enter the 2nd number: "))
num3=float(input("Enter the 3rd number: "))

if (num1 > num2) and (num1 > num3):
    largest=num1
elif (num2 > num1) and (num2 > num3):
    largest=num2
else:
    largest=num3

if (num1 < num2) and (num1 < num3):
    smallest=num1
elif (num2 < num1) and (num2 < num3):
    smallest=num2
else:
    smallest=num3

if (num1 < largest) and (num1 > smallest):
    mid=num1
elif (num2 < largest) and (num2 > smallest):
    mid=num2
else:
    mid=num3

print("smallest among 3 numbers is : ",smallest)

print("middle among 3 numbers is : ",mid)

print("largest number is: ",largest)
```